Aman Prakash

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Technical Skills

Languages	Python • TypeScript • JavaScript • Java • Rust • C++ • C • SQL • GraphQL • HTML • CSS
Frameworks	React • Flask • Django • Node • Express • LangChain • PyTorch • OpenCV • Espresso • JUnit
Platforms & Tools	AWS • GCP • Databricks • Datadog • Git/GitHub • Graphite • MySQL • MongoDB • PostgreSQL • Docker

Work Experience

Asana — *Vancouver, Canada*
Software Engineer — *May 2026 – PRESENT*

- Engineered a **3-stage data migration** by synchronizing legacy and greenfield backends, rerouting new writes, and executing a **shard-aware distributed migration job** that preserved **70K user-configured subscriptions** across production clusters
- Audited **82K subscriptions throughout staged rollout** using **Datadog** metrics, **Databricks** SQL queries, and live **JavaScript probes**
- Launched a **Microsoft Teams integration for Asana Gov** that uses asynchronous jobs to deliver **7 configurable project notifications** to connected conversations and lets users manage subscriptions, extending a product serving **40K active users across 70K projects**
- Spearheaded the design of **workspace-aware routing** for Microsoft Teams users, composing **data-flow and sequence diagrams** and building a **3-attempt backend resolver** with bounded backoff and origin-validated **authentication recovery**
- Investigated a data-model gap causing redundant API fan-out and designed a unified server endpoint with centralized logic that **reduced client calls by 67%**, enabled targeted retries, and avoided a complex backfill

Asana — *Vancouver, Canada*
Software Engineering Intern — *May 2025 – August 2025*

- Optimized and **A/B-tested** a paid-tier upgrade-request flow across **80+ product surfaces**, **reducing 4 steps to 3** and **increasing upgrade requests by 367%, tier upgrades by 44%, and higher-tier ARR per organization by 28%,** with a **$483K ARR impact**
- Integrated core work management with premium strategic planning features using **React/TypeScript components**, client-server mutations, server-computed values, and **GraphQL projections** to address a **need cited by 26% of churned users**
- Prototyped a **bidirectional Asana-Discord integration** in **3 days**, automating role-gated and real-time updates via REST APIs and webhooks; **won 1st place at Asana's company-wide internal hackathon**

Amazon Web Services • Cloud Innovation Centre — *Vancouver, Canada*
Software Developer Intern — *May 2024 – April 2025*

- Pioneered an **early reusable GenAI platform at UBC**; launched a cyber teaching assistant **serving 100+ computer science students**, then adapted its architecture into a virtual patient simulator **used in pharmacy classes of 75+ students**
- Upgraded architecture to support a student advising assistant that used **retrieval-augmented generation (RAG)** for course and policy inquiries; **presented the solution to UBC's President**, helping advance plans for **university-wide deployment**
- Replaced an always-on, self-hosted LLM with serverless **Amazon Bedrock**, **saving $4,897 USD** annually **while increasing model-request capacity by 16x** and **reducing text-generation latency by 85%** per query
- Architected a **serverless, event-driven AWS application** with **20+ Lambda functions** and **~60 OpenAPI operations**; automated repeatable deployments using **AWS CDK** across **API Gateway, SQS, AppSync, Cognito, RDS, DynamoDB, S3, ECS,** and **ECR**
- Built **LangChain RAG pipelines** that ingested documents from **S3 buckets**, generated **Amazon Titan embeddings**, and used **HNSW vector indexes** and **KNN search** in **PostgreSQL** with **pgvector** for **low-latency semantic retrieval** across content and metadata
- Productionized workloads with **multi-AZ RDS, RDS Proxy connection pooling, private subnets, VPC endpoints, WAF, Cognito,** and least-privilege **IAM roles**; implemented **SQS/AppSync asynchronous exports** and **CloudWatch/EventBridge** failure handling

Education

University of British Columbia — *Vancouver, Canada*
Bachelor of Applied Science - Computer Engineering — *Sept 2019 – May 2026*

- Relevant Coursework: Software Engineering (A), Software Testing & Analysis (A), Capstone Design Project (A)

Technical Projects

Object Identification via UAV Camera
Python, PyTorch, OpenCV, Google Cloud Vertex AI, Artifact Registry, Docker, JupyterLab, RF-DETR, ByteTrack — *April 2026*

- Created an onboard UAV perception system that **detects**, **tracks**, and **estimates vehicle speeds** across RGB/infrared video
- Orchestrated containerized model training on **Google Cloud Platform**, processing **13+ GB of data** and achieving **0.87 mAP@0.50**
- Applied **meter-per-pixel geometry, Lucas-Kanade optical flow**, and **ego-motion vector compensation** for real-time speed estimation